



Fig. 5: MEMS devices have changed the implementation of IMU functions, such as gyroscopes and accelerometers

outs. Its sensitive front-end requires signal strength of -159.5dBm for operation. After it computes locations based on GPS received signals, it provides output data to the system processor via the serial interface using industry-standard National Marine Electronics Association message format.

While GPS is an essential function for autonomous vehicles, it is not sufficient by itself. The GPS signal is blocked by canyons, tunnels, radio interferences and other factors, and these outages can last for many minutes. To supplement the GPS, the autonomous vehicle uses inertial guidance that requires no external signal of any type. The inertial measurement unit (IMU) consists of a platform fixed to the vehicle, and this platform has three gyroscopes and three accelerometers, one pair oriented each for orthogonal x, y and z axes. These sensors provide data on the rotational and linear motion of the platform, which is then used to calculate the motion and position of the vehicle, regardless of speed or any sort of signal obstruction. Note that, an IMU cannot tell you where you are, only the motion, so the initial location of the vehicle must be determined by the GPS or entered manually.

The in-vehicle IMU would not be practical without the development of MEMS based gyroscopes and accelerometers. The historical and fully-refined IMU is based on

a spinning-wheel gyroscope and a gimballed platform, which has served many applications quite well (missile guidance/Space missions), but it is simply too large, costly and power-hungry for an autonomous vehicle.

A representative MEMS device is A3G4250D IC from ST Microelectronics, a low-power three-axis angular rate sensor that provides a high degree of stability at zero rate level and with high sensitivity over temperature and time, shown in Fig. 5. It provides 16-bit digitised sensor information to the user's microprocessor via a standard SPI or I2C digital interface, depending on the chosen version. With its tiny size of just 4mm x 4mm, operation from a 1.8V supply, and stability and accuracy specifications, it is well-suited for inertial automotive navigation when combined with a three-axis accelerometer, for a complete six-axis IMU.

The autonomous car must be able to see and interpret what is in front when going forwards (and behind when in reverse, of course). It is also necessary to see what is on either side. In other words, it needs a 360-degree view. An array of video cameras is the obvious choice, plus a camera to determine where the lane is and sense objects or markers on the road.

But using cameras alone presents problems. First, there are mechanical issues of setting up multiple cameras

correctly and keeping these clean. Second, heavy graphic processing is needed to make sense of images. Third, there is a need for depth perception as well as basic imaging. And finally, conditions of lighting, shadows and other factors make it very challenging to accurately decide what the camera is seeing.

Instead, the primary vision unit on the autonomous vehicle is a lidar system. To enable the split-second decision-making needed for self-driving cars, a lidar system provides accurate 3D information on the surrounding environment. Using this data, the processor implements object identification, motion vector determination, collision prediction and avoidance strategies. Lidar unit is well-suited to big-picture imaging, and provides the needed 360-degree view by using a rotating, scanning mirror assembly on the top of the car.

Lidar provides raw information using high-speed, high-power pulses of laser light that are timed with the response of a detector to calculate the distance to an object from the reflected light. An array of detectors, or a timed camera, can be used to increase the resolution of the 3D information. The pulse is very short to enhance depth resolution, and the resulting light reflections are used to create a 3D point-like cloud that is analysed to transform the data into volume identification and vector information.

The transformed result is then used to calculate the vehicle's position, speed and direction, relative to these external objects, to determine the probability of collision and instruct appropriate action, if needed.

For close-in control, such as when parking, lane-changing or in bumper-to-bumper traffic, lidar system is not as effective. Therefore it is supplemented by radars built into the front and rear bumpers, and sides of the vehicle. Operating frequency for this radar is usually 77GHz, which has been allocated for this use, has good RF propagation characteristics and